



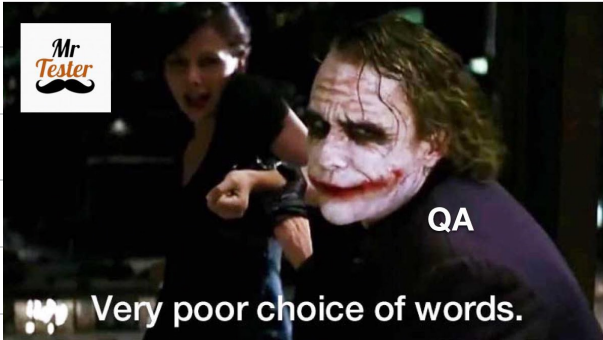
AGENDA

- ① What is Docker??
- ② Diff b/w Docker, Virtual machine and virtual environments.
- ③ Workflow of Docker
- ④ Docker hub

WHAT IS DOCKER

WHY WE NEED IT?

Dev: Works in my machine...



① Python version
2.7

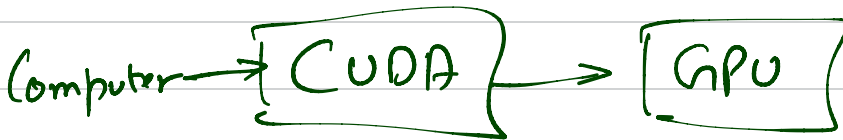
② Database

③ Frontend

④ Backend = Spring

④ CUDA → Fickir
Python
+ tensorflow

We need somethin else??



→ Graphical Processing Unit

→ Platform → building, running & shipping
applications → conveniently

Python + database + Frontend
isolated environment.

↓
CONTAINER



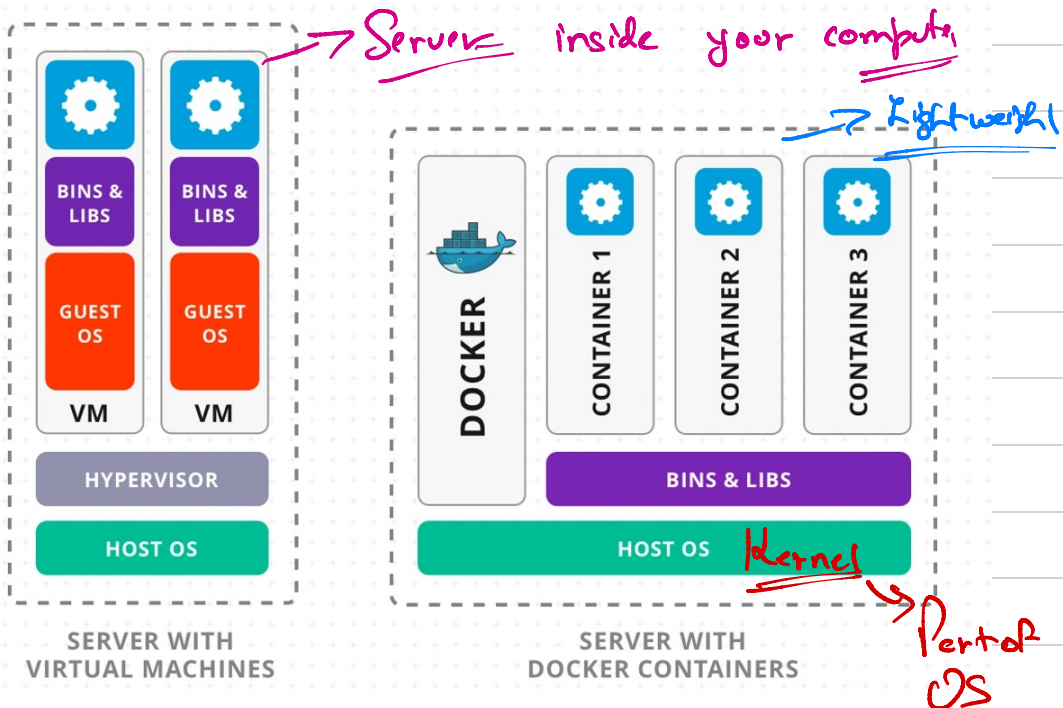
→ primary reasons for issues:

* Compatibility issue → Windows
vs
Linux

* Missing Software

* Incapable Hardware??

Virtual Machine vs docker



that controls

16 GB Ram
1 TB → HDD

RAM, HDD
Vm does
quarantine → 30%
of your RAM
+
HDD

→ Hypervisor → reserves

Resources For simulating
actual hardware For VM

Simulate on actual
hardware → for
the Virtual machine

→ Docker

- (i) V. Light weight
- (ii) Ram is consumed as needed

Not quarantine

(can specify limit) → some - RAM
- HDD

- (iii) Little less secure than VM

→ Windows → WSL (Windows 10 →)

WSL → lets docker simulate
behave
Like Linux

Problems, when to use what?

① Docker → Host Kernel

root → superuser → all the rights

→ Neo (not root)



Creates → docker container
on my system

→ delete *

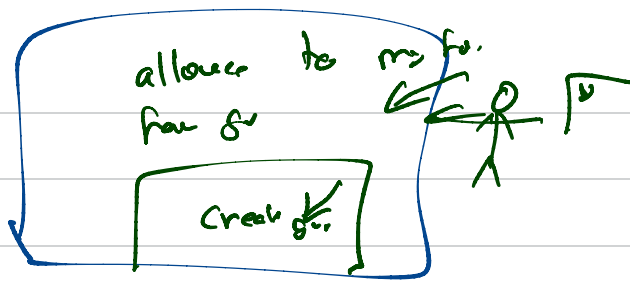
→ cp *

→ rm *

super-admin

Admin / root of my docker container

access my computer
as root



namespaces
multiple level
of user
access

→ Deployment

- ① Find some cloud environment (AWS, GCP, Azure, digital ocean)
- ② Install some OS → Linux, ~~Windows~~
- ③ Install dependencies / Libraries.
- ④ Copy the code → Repo → Server
- ⑤ Run the code

DOCKER

① Docker File → Recipe

② Image → Pre made Food → Fridge

③ Container → Serving to Users

Image / package in motion